L1: Algorithmic trading basics and financial markets

Course Code: COMP7415

Lecturer: Tony Lam

Agenda

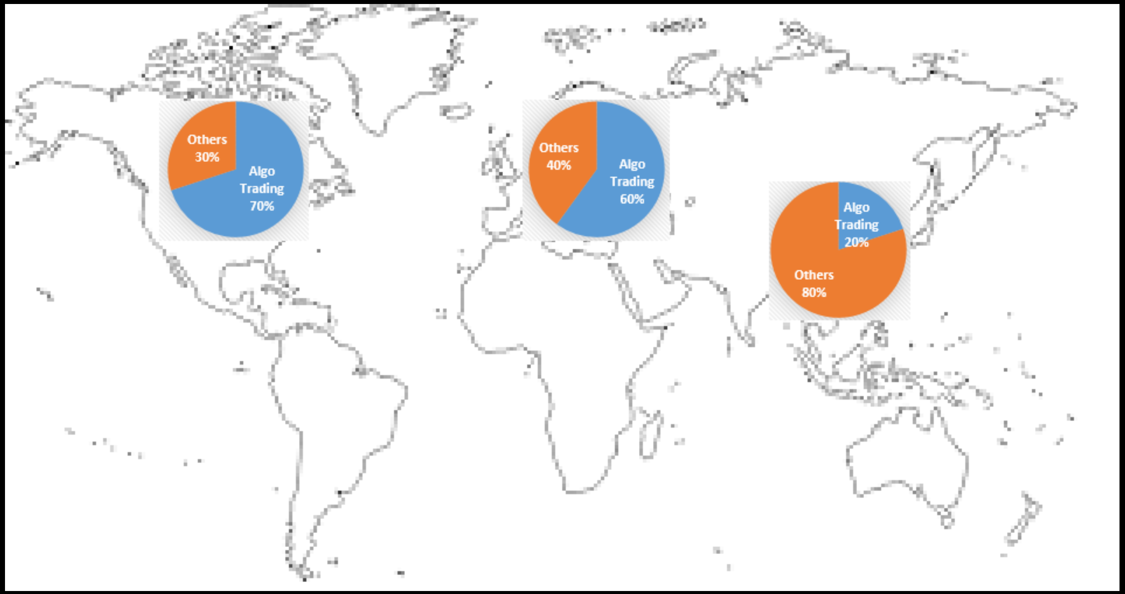
- The basic concept of algo-trading
- The market trend of algo-trading
- Comparison of different trading approaches
- Infrastructure needed to deploy algo-trading programs
- Trading mechanism and order matching
- Market jargon and terminology

What is algo-trading?

- Algorithmic trading, also called Algo Trading, Quant Trading, Robo-Trading, Program Trading
- Use of mathematical models and computer algorithms/programs to
  - generate trading signals (i.e. **Decision making**)
  - automate trading process (i.e. **Execution**)
- Objectives:
  - maximize profits
  - control execution costs
  - hedging and manage investment risks

Market Trend for Algo-Trading

- Expected to grow at CAGR 12.7% to US$32b by 2028
- Largest market in North America
- Fastest growing in Asia Pacific



Reference:
https://www.mordorintelligence.com/industry-reports/algorithmic-trading-market
https://www.prnewswire.com/in/news-releases/algorithmic-trading-market-to-reach-usd-31-494-million-by-2028-at-a-cagr-of-12-7-valuates-reports-897347529.html

Market Trend for Algo-Trading

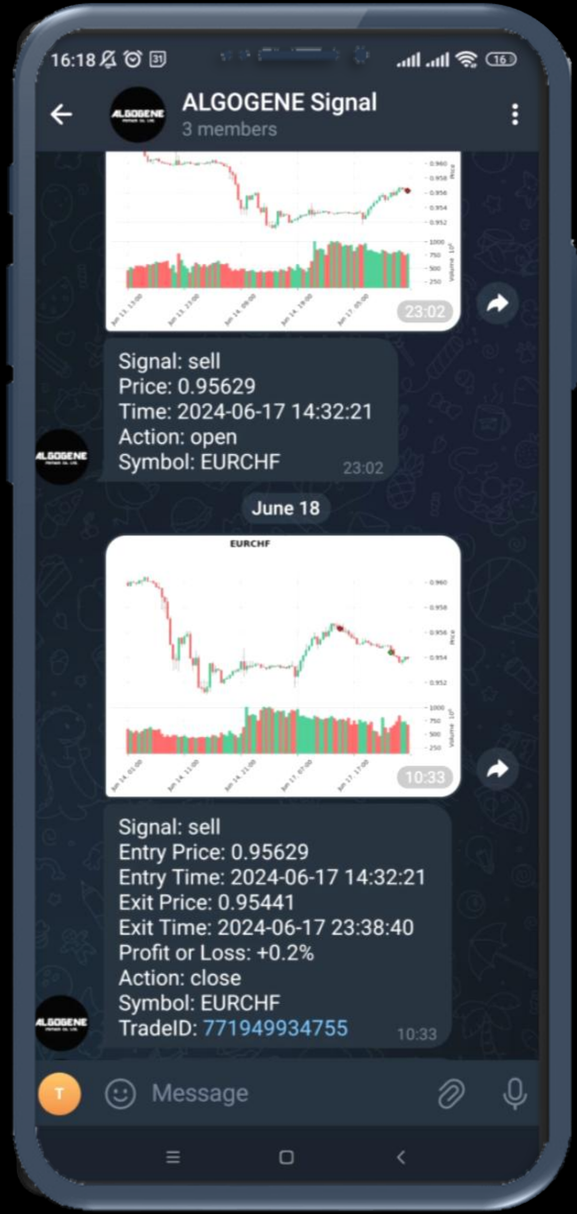
Algorithmic trading strategies account for approximately:

- 60% of all US equity volume
- 40% of all European equity volume
- 25% of all Forex transactions
- 20% of all US option trades

Source: interactive brokers

Robo-Trading vs Robo-Advisory

	Robo-Trading	Robo-Advisory
Decision making	✓	✓
Execution	✓	╳
Automation	Fully automated	Semi-automated
Outputs	Execution of trading signals	• Stock advices • Market summary • Risk alerts



Telegram Group: https://t.me/algogene_signal

Comparison of Trading Approaches

manual ⟷ automatic

	Human Trading	Robo-Advisory	Robo-Trading
Working Hours	~9*5	~9*5	24*7
Execution Speed	Slow	Moderate	Immediately
Data Inputs	Limited	Almost unlimited	Almost unlimited
Trading Frequency	Low	Low - mid	Low – High
Scalability	Limited investment size due to stress	Moderate	High
Accuracy/ Discipline	Variable especially when you lose	Consistent	High
Customization	High	Medium	Low
User Control	Full	Partial	Minimal
Risk Management	Subjective	Algorithmic	Algorithmic

What should you consider to implement a trading algo?

Algo is the vehicle; Data is the Fuel!

- **Data collection** – time and effort demanding in data sourcing and data cleaning
- **Data storage** – high volume of unstructured data (Numeric, Text, Image, Audio, Video)
- **Data management** – difficult to protect and manage through traditional databases due to inconsistent file and data formats
- **Data processing** – feasibility and efficiency

Other factors for infrastructure setup

- Real-time datafeed
- Historical data
- Computing hardware
- Exchange/broker connectivity
- Network and latencies
- Order routing system
- Order management system
- Algo maintenance
- Cybersecurity
- …

General Trading Overview

Types of Trading

Fundamental:
- Stock Selection
- Ratio Analysis
- Sector Analysis
- Corporate and Executive Management

Technical:
- Chart pattern
- Trend Analysis

Quantitative:
- Rule based
- Econometric Forecasting
- Statistical Arbitrage
- NLP, sentiment analysis
- Machine learning

Types of Trading

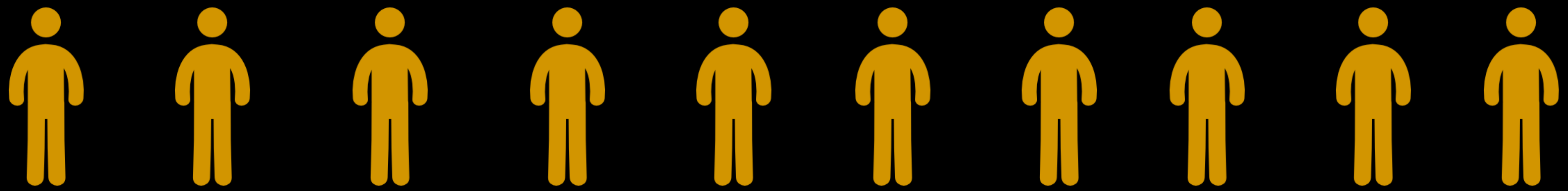
Time Frames:

- Long Term: Months to Years
- Short Term: Days, Weeks, Months
- Intraday: Seconds to Hours
- High frequency: Fractions of Seconds

How trading works?

Supply

Current Stock holders
(each person holds 1 share)

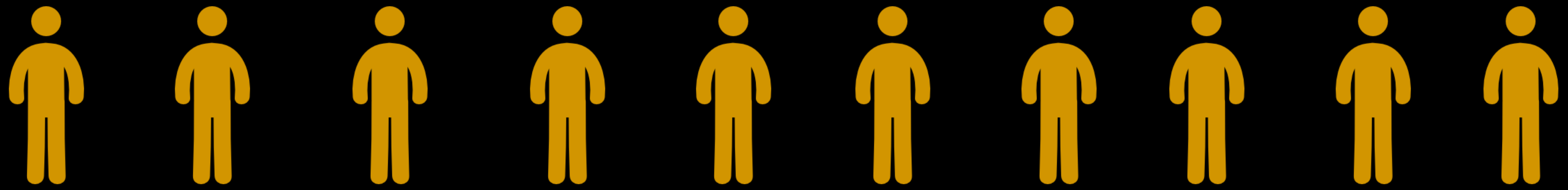


$10 $12 $13 $15 $15 $18 $21 $24 $25 $30

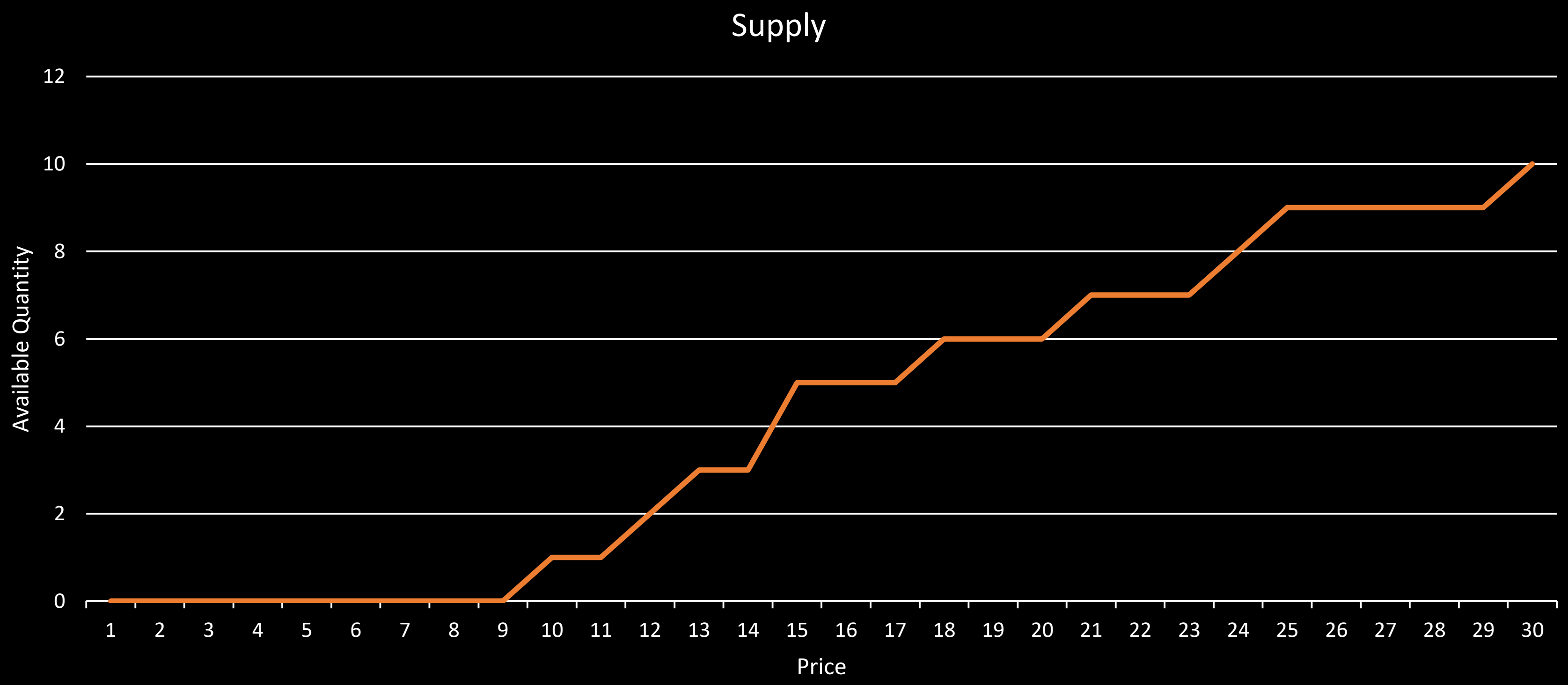
Price they would be willing to take (sell) for their shares

Current Stock holders

(each person holds 1 share)



$10 $12 $13 $15 $15 $18 $21 $24 $25 $30

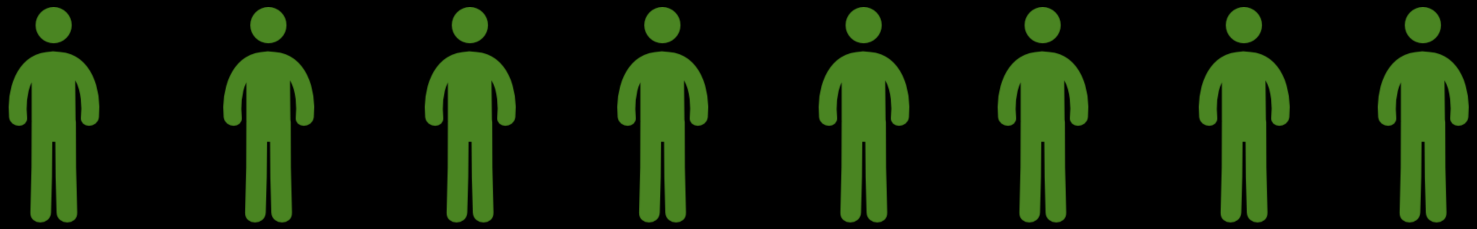


Price	Supply
1	0
2	0
3	0
4	0
5	0
6	0
7	0
8	0
9	0
10	1
11	1
12	2
13	3
14	3
15	5
16	5
17	5
18	6
19	6
20	6
21	7
22	7
23	7
24	8
25	9
26	9
27	9
28	9
29	9
30	10

Demand

Potential Buyers

(each person wants to buy 1 share)

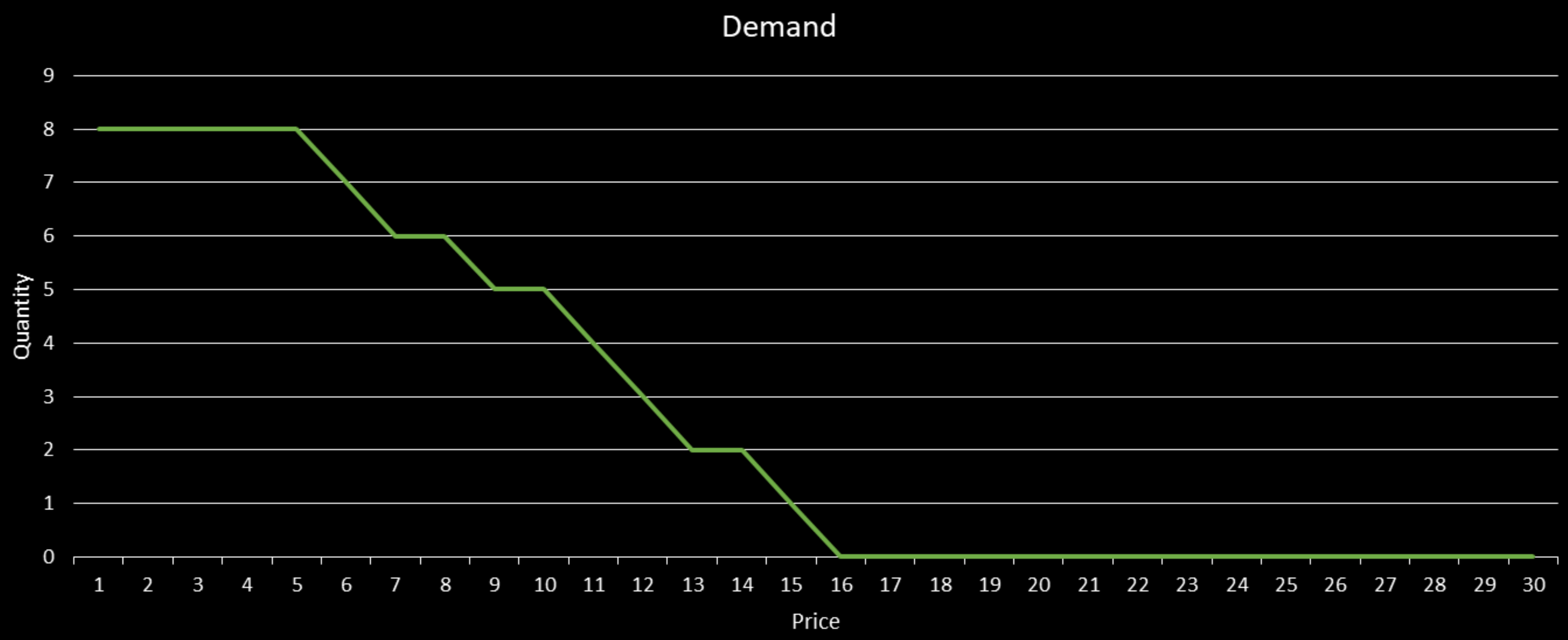


$5 $6 $8 $10 $11 $12 $14 $15

Price they would be willing to pay for a share

Potential Buyers

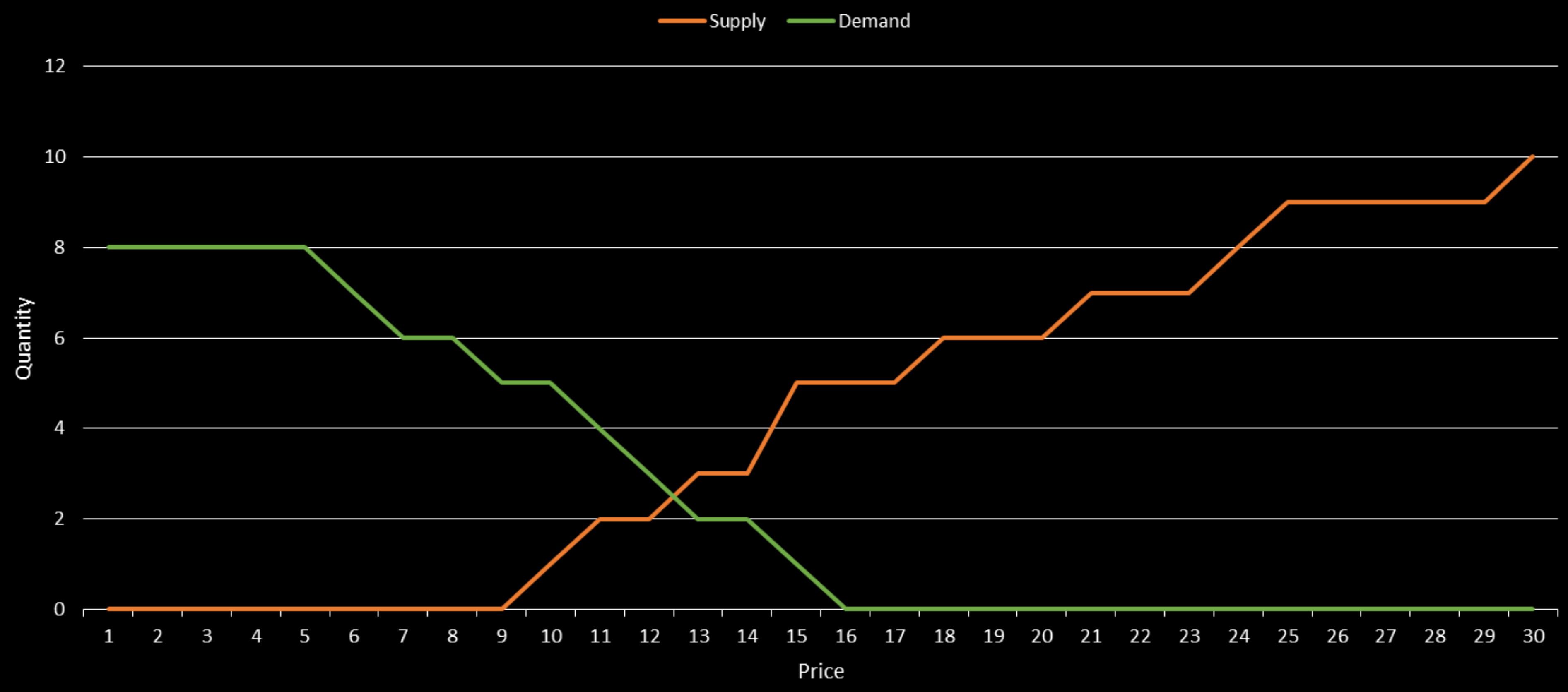
(each person wants to buy 1 share)



Price	Demand
1	8
2	8
3	8
4	8
5	8
6	7
7	6
8	6
9	5
10	5
11	4
12	3
13	2
14	2
15	1
16	0
17	0
18	0
19	0
20	0
21	0
22	0
23	0
24	0
25	0
26	0
27	0
28	0
29	0
30	0

Market Equilibrium

- The market is balanced at $12 and quantity 2



Price	Supply	Demand
1	0	8
2	0	8
3	0	8
4	0	8
5	0	8
6	0	7
7	0	6
8	0	6
9	0	5
10	1	5
11	1	4
12	2	3
13	3	2
14	3	2
15	5	1
16	5	0
17	5	0
18	6	0
19	6	0
20	6	0
21	7	0
22	7	0
23	7	0
24	8	0
25	9	0
26	9	0
27	9	0
28	9	0
29	9	0
30	10	0

Order Matching

Current Stock holders

(each person holds 1 share)

Potential Buyers

(each person wants to buy 1 share)

After Order Matching

Market Terminology

Common market terminology

1. Long and short position
2. Short-selling
3. Order book
4. Bid price, ask price
5. Market spread
6. Order types
7. Slippage
8. Mark-to-market
9. Trading system
10. Margin & leverage

Understanding Long and Short Positions

- Long Position
  - **Definition**: Buying an asset with the expectation that its price will rise.
  - **Goal**: Sell at a higher price to realize a profit.
  - **Example**: Buying 100 shares of a stock at $50, selling when it reaches $70.

Understanding Long and Short Positions

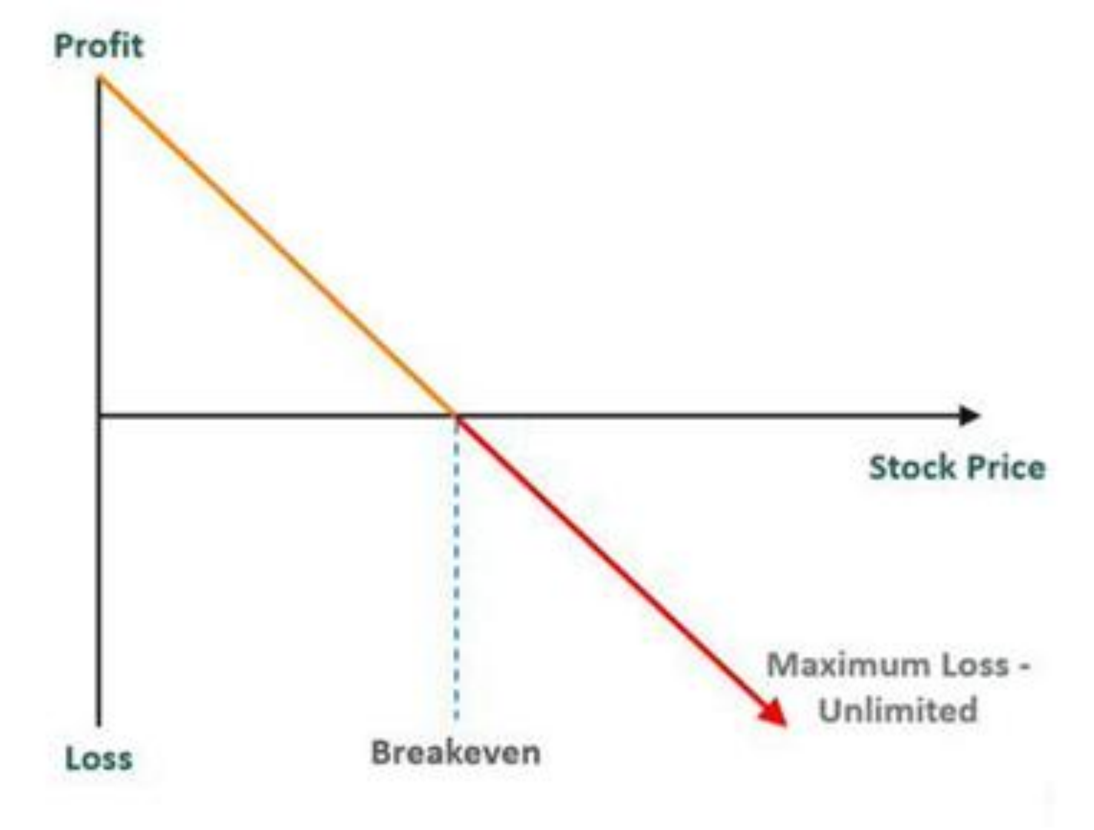
- Short Position
  - **Definition**: Selling an asset that you do not own, with the intention of buying it back later at a lower price.
  - **Goal**: Profit from a decline in the asset's price.
  - **Example**: Selling 100 shares of a stock at $70, buying back at $50.

How Short-Selling Works

1. Borrowing Shares: Obtain shares from a broker to sell.
2. Selling the Borrowed Shares: Sell the shares at the current market price.
3. Buying Back Shares: Later, buy the same number of shares at a lower price.
4. Returning Shares: Return the borrowed shares to the broker.

Risks of Short-Selling

- **Unlimited Loss Potential**: If the asset price rises significantly.
- **Margin Calls**: Brokers may require additional funds if the trade goes against you.
- **Regulatory Risks**: Short-selling can be subject to restrictions.



Order Book

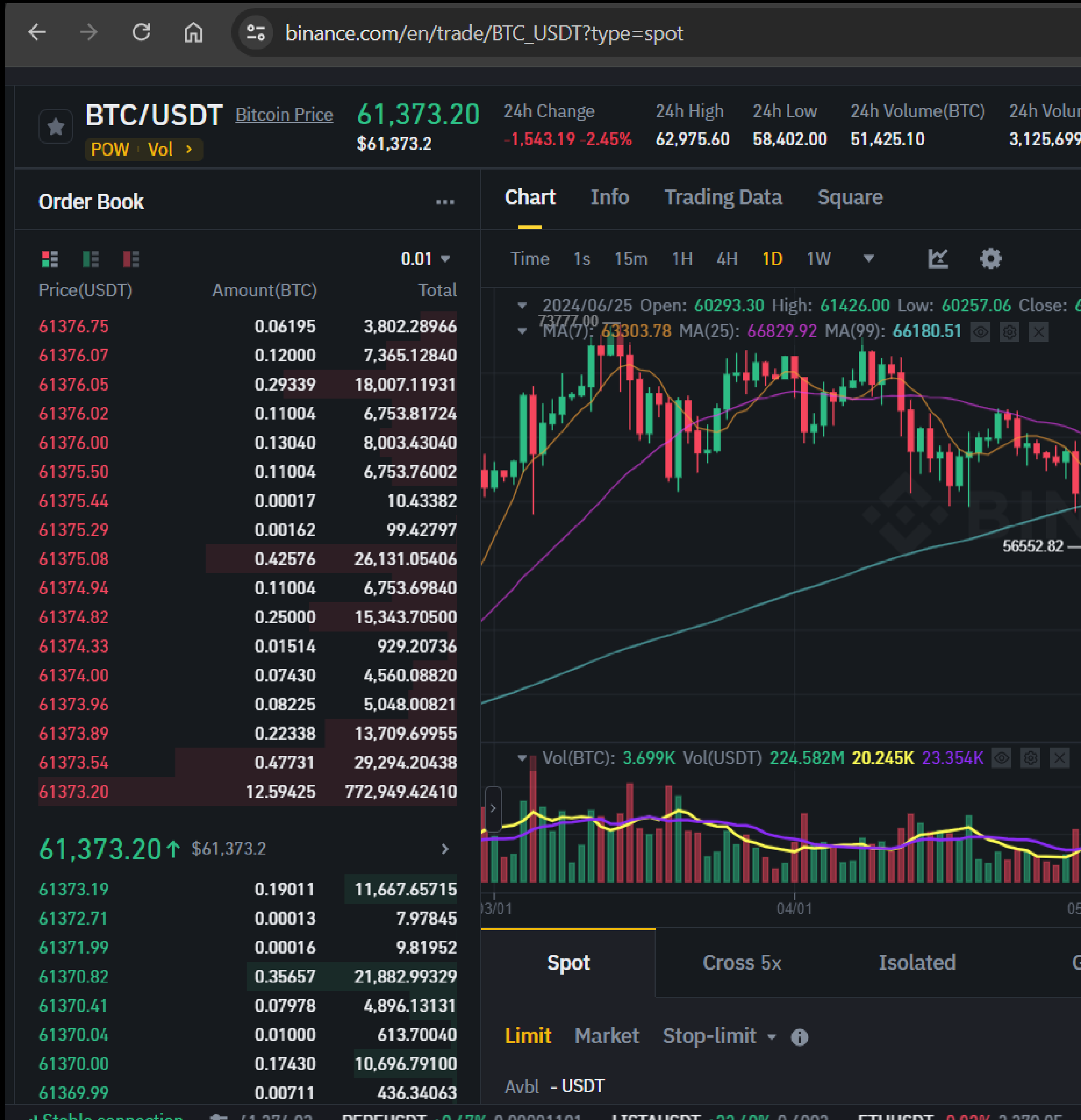
- The Order Book is a real-time, continually updating list of buy and sell orders for a specific financial instrument, sorted by price level.
- Key components
  - Ask Book: the list/queue for sellers
  - Bid Book: the list/queue for buyers
  - Price: The price points at which investors are willing to trade
  - Size: The number of shares, contracts, or lots that traders want to buy or sell

How does Order Book present?

1. Order Ledger
2. Top of the Book
3. Cumulative Book

Order Ledger

- The order book is listed in a sorted price ledger, which can be ascending or descending.
- The longer the price ledger is, the more liquidity an instrument provides. It is also called "market depth".



Top of the Book

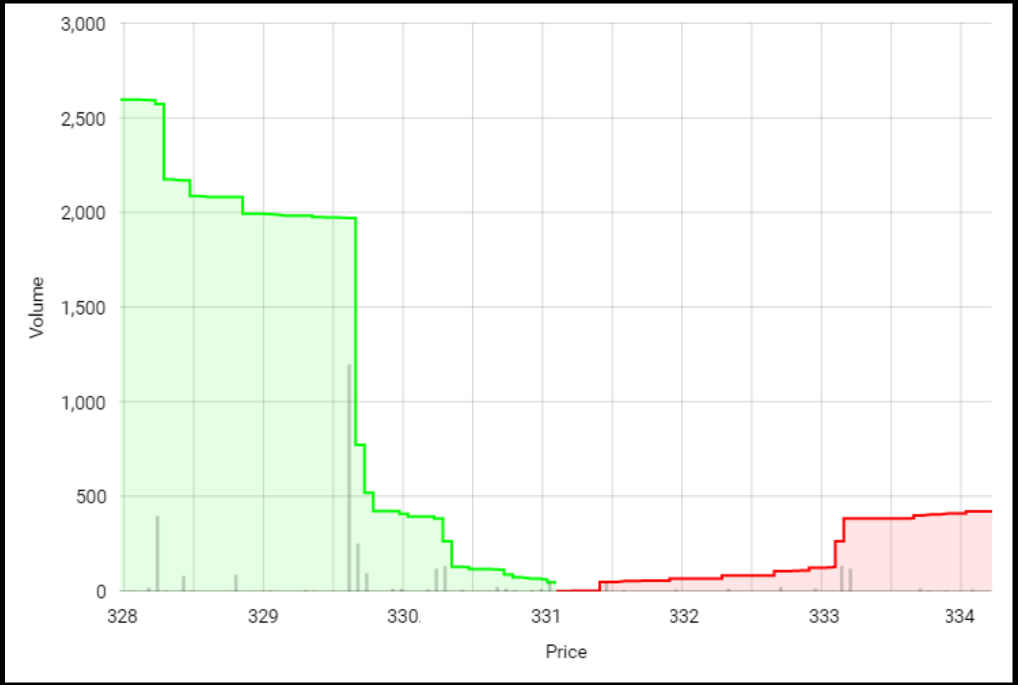
- The order book is separated into bid-book and ask-book
- Bid orders are sorted in descending price
- Ask orders are sorted in ascending price
- The highest bid and the lowest ask are referred to "the top of the book"

Order Book

	Bid	Ask	
3,151	9139.60	9152.80	167
1,574	9138.40	9153.21	1,600
338	9138.20	9153.60	736
789	9138.10	9153.71	1,602
800	9138.00	9154.10	1,649
309	9137.79	9154.21	1,965
2,140	9137.60	9154.50	3,984
48	9137.40	9154.60	763
144	9137.29	9154.71	157
136	9137.20	9154.80	56
528	9137.00	9155.50	320
1,654	9136.90	9155.71	200
40	9136.79	9156.10	16
46	9136.70	9156.21	219
608	9136.60	9156.30	40
48	9136.50	9156.40	80

Cumulative Order Book

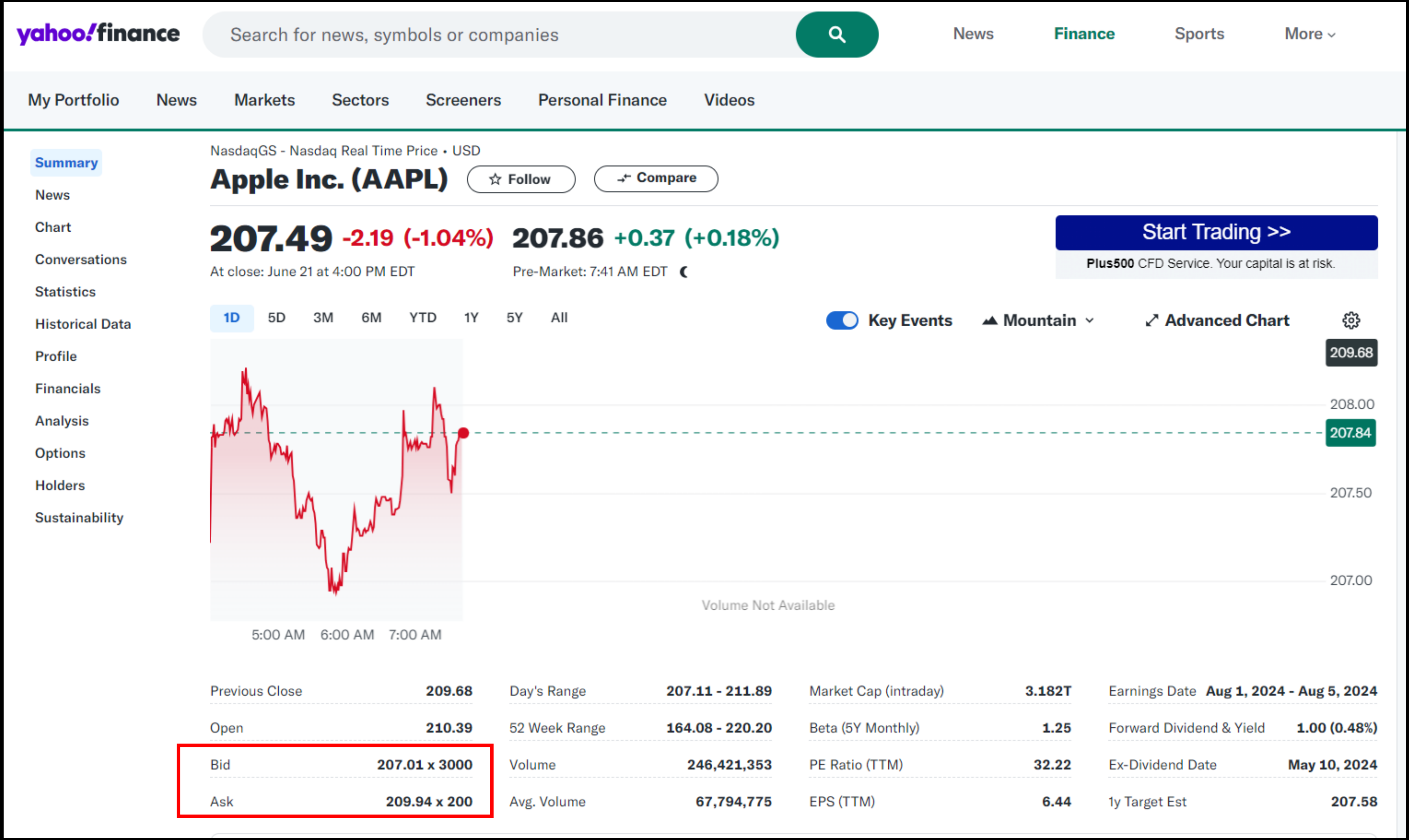
- For each of the bid and ask book, order size accumulates from the top of the book.



Order Book Market Trades

0.1

Price(USDT)	Qty(BTC)	Total(BTC)
61,284.8	11.972	54.521
61,284.7	7.282	42.549
61,284.0	2.529	35.267
61,283.9	1.305	32.738
61,283.8	1.795	31.433
61,283.7	4.569	29.638
61,283.5	5.773	25.069
61,283.4	1.795	19.296
61,283.3	3.590	17.501
61,283.2	13.911	13.911
61,284.4 ↑ 61,291.9		
61,282.3	12.075	12.075
61,282.2	0.816	12.891
61,282.1	2.693	15.584
61,282.0	4.712	20.296
61,281.9	9.444	29.740
61,281.8	1.795	31.535
61,281.7	3.672	35.207
61,281.6	5.181	40.388
61,278.4	15.301	55.689
61,267.2	15.306	70.995



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NasdaqGS - Nasdaq Real Time Price • USD

Apple Inc. (AAPL)

Follow Compare

207.49 -2.19 (-1.04%) **207.86** +0.37 (+0.18%)

At close: June 21 at 4:00 PM EDT Pre-Market: 7:41 AM EDT

Start Trading >>

Plus500 CFD Service. Your capital is at risk.

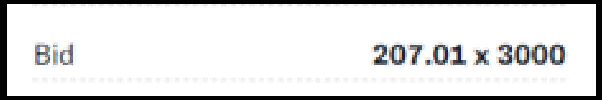
1D 5D 3M 6M YTD 1Y 5Y All

Key Events Mountain Advanced Chart

Previous Close	209.68	Day's Range	207.11 - 211.89	Market Cap (intraday)	3.182T	Earnings Date	Aug 1, 2024 - Aug 5, 2024
Open	210.39	52 Week Range	164.08 - 220.20	Beta (5Y Monthly)	1.25	Forward Dividend & Yield	1.00 (0.48%)
Bid	207.01 x 3000	Volume	246,421,353	PE Ratio (TTM)	32.22	Ex-Dividend Date	May 10, 2024
Ask	209.94 x 200	Avg. Volume	67,794,775	EPS (TTM)	6.44	1y Target Est	207.58

Bid Price & Bid Size

- Bid price is the highest price that a buyer (bidder) is willing to pay for a particular security.
- Bid size represents the quantity a buyer is willing to purchase at the bid price.
- The top price and size in a bid order book.
- For example, if a buyer bids $207.01 x 3000 for a stock, it means they are willing to buy 3000 shares for not more than $207.01



Ask Price & Ask Size

- Ask price is the lowest price a seller is willing to accept for a security.
- Ask size represents the quantity a seller is willing to sell at the ask price.
- The top price and size in an ask order book.
- For example, if a seller asks $209.94 x 200 for a stock, it implies they are willing to sell 200 shares for not less than $209.94

Ask	209.94 x 200

HKEX Market Data Pricing

- Level 1 data:
  - bid/ask price: HK$500 /month
- Level 2 data:
  - full order book: HK$5,000 /month

HKEX
香港交易所

Historical Full Book - Securities Market (Binary Format) provides information on every order and trade of Main Board and GEM stocks in binary format. File is available on a daily basis, normally at around 8:30pm.	Daily	$5,000/ Month
Historical Full Book - Securities Market (CSV Format) provides information on every order and trade of Main Board and GEM stocks in CSV format. File is available on a daily basis, normally at around 10:00pm.	Daily	$5,000/ Month
Historical Order Book and Statistics Update - Securities Market provides intra-day order book and statistics information recorded on the HKEX's Main Board and GEM stocks in CSV format. File is available on a daily basis, normally at around 10:00pm.	Daily	$2,500/ Month
Bid and Ask Record - Main Board & GEM (Daily File) provides intra-day bid and ask information on Main Board and GEM stocks	Daily	$500/ Month
Bid and Ask Record - Main Board & GEM (Monthly File) provides intra-day bid and ask information on Main Board and GEM stocks	Monthly	$500/ Month
Trade File - Securities Market (Binary Format) provides trade details, including trade time, price and quantity, of each transaction recorded on the Main Board and GEM in binary format. File is available on a daily basis, normally at around 8:30pm.	Daily	$2,500/ Month
Trade File - Securities Market (CSV) provides trade details, including trade time, price and quantity, of each transaction recorded on the Main Board and GEM in CSV format. File is available on a daily basis, normally at around 10:00pm.	Daily	$2,500/ Month

Reference: https://www.hkex.com.hk/eng/ods/historicalData.aspx

Market Spread

- The difference between the bid price and the ask price is known as the bid-ask spread.
- The bid-ask spread is a key measure of the liquidity of an asset or security.
- A smaller spread indicates a more liquid market, while a larger spread indicates a less liquid market.
- For example, if the bid price for a stock is $24 and the ask price is $25, the bid-ask spread will be $1.

Percentage Spread

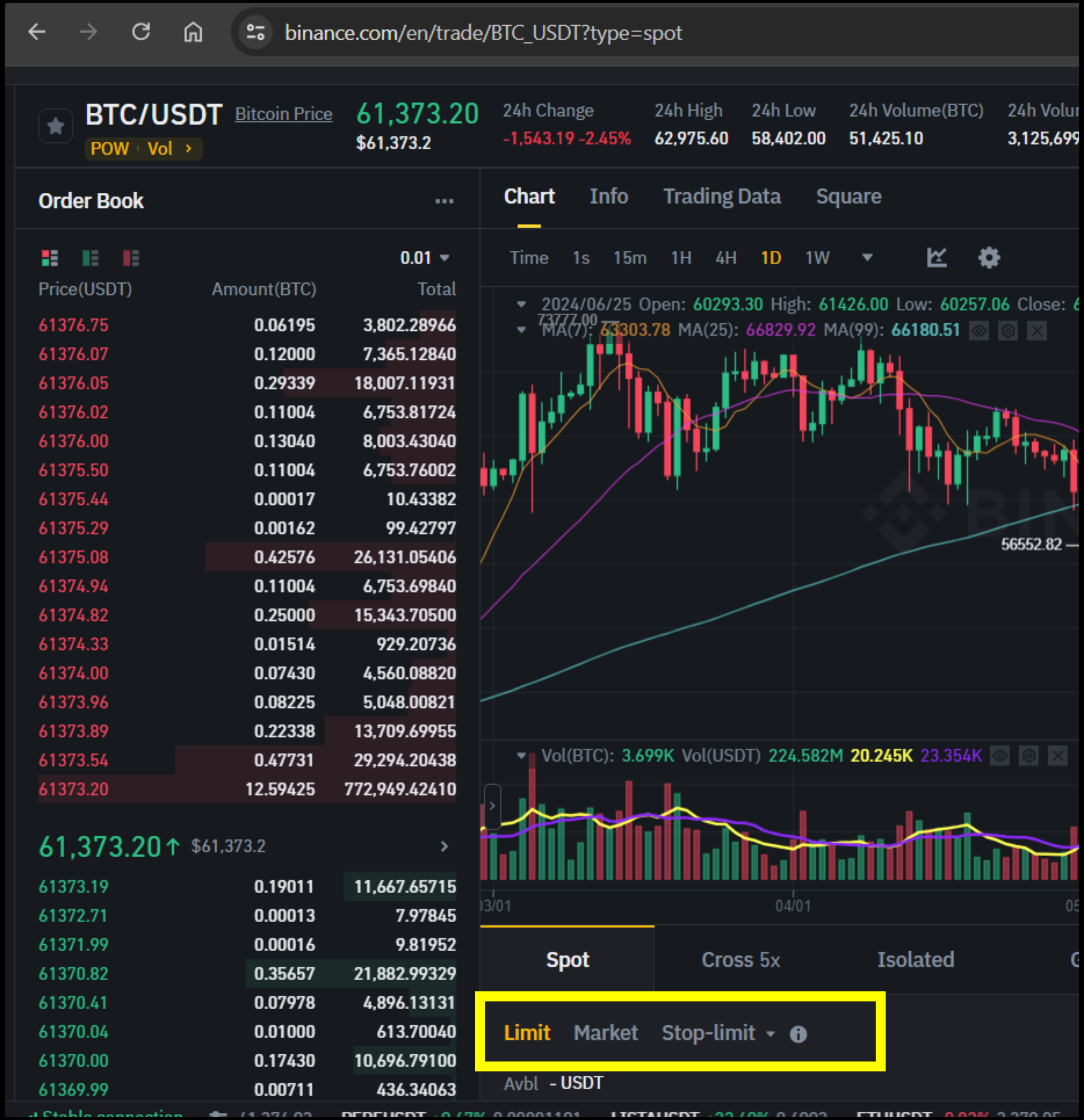
- Percentage spread is often used for comparing the spread between different instruments

$$Percentage\ Spread = \frac{AskPrice\ - BidPrice}{MidPrice} \times 100\% = \frac{2 \times (AskPrice\ - BidPrice)}{AskPrice + BidPrice} \times 100\%$$

- For example, if the bid price for a stock is $24 and the ask price is $25,
  - the bid-ask spread will be (25-24) = $1
  - the percentage spread will be 2*(25-20)/(25+20) = 4.08%

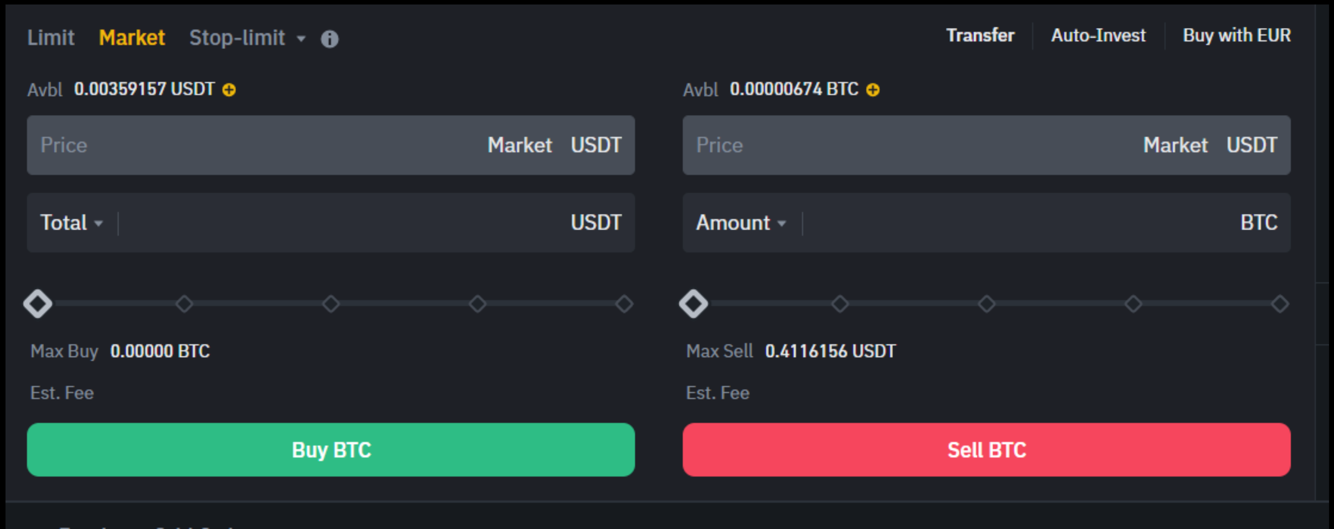
Order Types

1. Market Order
2. Limit Order
3. Stop Order



Market Order

- A market order is to buy or sell at the best available price in the current market.
- That is, when opening a buy (sell) position using market order, it will try to execute at the best ask (bid) price.
- A market order typically ensures an execution, but it does not guarantee a specified price. It is appropriate to use market orders when you want an immediate execution.



Limit Order

- A limit order is to buy or sell with a condition on the maximum price to pay or the minimum price to receive (the "limit price").
- If the order is filled, it will only be at the specified limit price or better. However, there is no guarantee of execution.
- A limit order may be appropriate when you think you can buy at a price lower than (or sell at a price higher than) the current market quote.

Limit Order Example

Limit Order Example

- The last trade price is roughly $138
  - investor who wants to buy (or sell) immediately would place a market order, which would be executed at or near the current price of $138 (white line) provided that the market was open.
  - investor who wants to buy the stock when it dropped to $131.78 would place a buy limit order with a limit price of $131.78 (green line). If the price falls to $131.78 or lower, the limit order would be triggered and executed at $131.78 or below. If the stock doesn't drop to $131.78 or below, no execution would occur.
  - investor who wants to sell the stock when it reached $143.82 would place a sell limit order with a limit price of $143.82 (red line). If the price rises to $143.82 or higher, the limit order would be triggered and executed at $143.82 or above. If the stock doesn't rise to $143.82 or above, no execution would occur.

Stop Order

- A stop order is to buy or sell at the market price when it has traded at or through a specified price (the "stop price").
- If the stock reaches the stop price, the order becomes a market order and is filled at the next available market price. If it doesn't reach the stop price, the order will not be executed.
- A stop order may be appropriate in these scenarios:
  - you want to buy when a stock breaks out above a certain level, and believe that it will continue the trend
  - your holding stock has risen a lot and you want to protect the gain when it begins to fall

Stop Order Example

Stop Order Example

- The stop buy order will trigger when the price reaches 143.82 or above, and will execute as a market order at that current price.
- Thus, if the price rise further after hitting the stop price, it is possible that the order could be executed at a price higher than the stop price.
- Similarly for the stop sell order, once the stop price of $131.78 is reached, the order could be executed at a lower price

Slippage

- Slippage refers to the difference between the expected execution price and the price it actually traded
- It often occurs
  - during periods of higher volatility when market orders are used
  - when large orders are executed when market depth is not sufficient to maintain the expected price of trade

Slippage Example

- Let's assume you have an algorithm set to buy a stock when the price drops to $50. The algorithm detects the price drop and places a buy order.
- However, due to high demand or low liquidity, the order gets executed at $51. This $1 difference is the slippage.
- This means that even if you were expecting to spend $5000 on 100 shares, you would end up spending $5100. The $100 is the cost of slippage

Profit/Loss Calculation

- **Entry**: Based on a trade signal, generate order to go short or long a certain financial instrument in a certain quantity. Trade results in a certain position in this security
- **Mark-to-Market**: As the price of the security changes, so does your **unrealized PnL**

For long order, PnL = Quantity $* (P_{bid} - P_{entry})$

For short order, PnL = $-1 *$ *Quantity* $* (P_{ask} - P_{entry})$

- **Exit**: Generate order to exit the position and create a **realized PnL**

PnL = Side $*$ Quantity $* (P_{exit} - P_{entry})$

Mark-to-Market

- It is an accounting method to calculate the asset or portfolio value according to its current market value, rather than its book value
- It involves determining the price at which you could immediately sell an asset or close a position
- Example:
  - Suppose you purchase 100 shares of ABC stock
  - The current bid and ask price of ABC are $9.5 and $10.5
  - Mark-to-market value of your stock will be 100*$9.5 = $9500

Order Management System (OMS)

1. Order based system
    - Transactions are managed in a round order manner
    - Partial close is not supported
    - Trading platforms
        - MetaTrader, TradingView, MetaStock, etc
2. Position based system
    - Transactions are independent and no linkage with each other
    - Partial close is allowed
    - Trading Platforms
        - Most of the banks (eg. HSBC, SBC, BOC, etc)
        - Interactive Brokers, Binance, etc

Position Based vs Order Based System

Suppose you open 3 offsetting trades:

#1: buy 2 shares of ABC at $123

#2: sell 1 share of ABC at $124

#3: sell 1 share of ABC at $125

For order based system, the **unrealized** PnL will be 2*(bid - 123) + 1*(124 - ask) + 1*(125 - ask) = $3 + 2*(bid - ask). You need to close all the 3 trades to realize the PnL.

For position based system, the orders will be offsetted based on FIFO, and PL will become realized so long as the position is net to zero. Thus, your PnL will be **realized** at $3.

Leverage & Margin

Leverage & Margin

- Leverage refers to using borrowed capital as a funding source for investment to increase the potential return

Leverage Ratio = Asset/ Equity = 1 + Debt/Equity

- Margin is a way to create leverage

Margin Amount = (Account Asset Value) - (Borrowed Amount) = Equity held at broker

Margin Requirement

There are 2 market quotations for margin requirement

1. **Fixed Margin Amount**: Mostly used by stock exchanges
2. **Margin as a percentage**: Mostly used by FX/Crypto exchanges

Update Date : 20251006

Please be reminded that the minimum margin rates below are for your firm's financially strongest clients.
Exchange Participants should set their margin requirements according to each client's individual circumstances.

Index Futures

Effective Date	Product	HKATS Code		Client Margin Initial (HK$)	Client Margin Maintenance (HK$)	Clearing House Margin (HK$)
20251002	Hang Seng Index	HSI	Full Rate /lot	116,774	93,419	87,800
			Spread Rate /spread	25,735	20,588	19,350
20251002	Mini - Hang Seng Index	MHI	Full Rate /lot	23,354	18,683	17,560
			Spread Rate /spread	5,147	4,117	3,870

Reference: https://www.hkex.com.hk/Services/Clearing/Listed-Derivatives/Risk-Management/Margin/Margin-Tables?sc_lang=en

How Margin is related to Leverage?

- As HSI Index Future has a price magnifier of 50, meaning that 1 index point change will lead to ±HK$50. Suppose current HSI Index Future price is 25000, and buying 1 lot of HSI Index Future require HK$116,774 as margin. Then, leverage ratio is calculated to be ($25000*50)/$116,774 = 10.7044
- Suppose you pay $600,000 to get stocks worth $1,000,000. Thus, leverage ratio is calculated to be $1,000,000/$600,000 = 1.6667
- As you can imagine, the lower is the margin requirement, the higher is the leverage ratio. In general,

$$Leverage\ Ratio = \frac{1}{Margin\ Requirement}$$

Buying Power

- For leveraged trading where traders can take out a loan based on the amount of cash held in their broker account, buying power refers to the amount of money available for investors to purchase securities. Mathematically,

$$\textit{Buying Power} = (\textit{Leverage Ratio}) \times (\textit{Investor's equity})$$

- For example, suppose an investor made an initial deposit US$100,000 to a 20:1 leveraged broker account. Then, the investor would be able to purchase, at maximum, US$2,000,000 worth of securities. Hence, a buying power of US$2,000,000

Key Takeaways

- Learn the basic concept of algo-trading and its market trend
- Understand the pros and cons of different trading approaches
- Know about the infrastructure/resources needed to develop and deploy algo-trading programs in real market
- Understand the trading mechanism and order matching in financial market
- Equip the knowledge about different market terminology (eg. Order book, order types, market spread, slippage, order/position based trading system, margin, etc)